IGCSE 0607 Practice Questions — E9.1 — Combined Questions & Markscheme

Introduction to probability

Total Questions: 15 — **Total Marks:** 93 Marks

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  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

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1 Event A is that a train arrives on time.

$P(A) = 0.83$

(a) Work out $P(A')$.

..... **[1]**

(b) State in words what the event A' represents.

..... **[1]**

2 A bag contains only red, blue, and green marbles.
The probability of picking a red marble is $\frac{2}{7}$ and the probability of picking a blue marble is $\frac{1}{3}$.

(a) Find the probability of picking a green marble.

..... **[2]**

(b) Let R be the event of picking a red marble.
Write down the value of $P(R')$.

$P(R') =$ ..... **[1]**

3 A biased spinner can land on 1, 2, 3, 4, or 5.
The table shows the probabilities of the spinner landing on each of the numbers.

Outcome	1	2	3	4	5
Probability	0.15	$2x$	0.31	x	0.24

(a) Find the value of x.

$x =$ ..... **[2]**

(b) The spinner is spun once.
Find the probability that it does not land on 2.

..... **[2]**

4 A biased spinner can land on Red, Blue, Green, or Yellow.
The table shows the probabilities of the spinner landing on Red, Blue, and Green.

Colour	Red	Blue	Green	Yellow
Probability	0.15	0.3	0.45	

(a) Work out the probability that the spinner lands on Yellow.

..... **[2]**

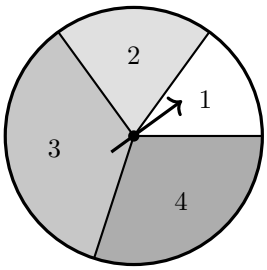
(b) Work out the probability that the spinner lands on Red or Blue.

..... **[1]**

(c) Let G be the event that the spinner lands on Green.
Find $P(G')$.

$P(G') =$ ..... **[2]**

5 The diagram shows a biased 4-sided spinner with sectors numbered 1, 2, 3, and 4.
When the spinner is spun once, the probabilities of landing on each number are $P(1) = 0.15$, $P(2) = 2x$, $P(3) = 0.35$, and $P(4) = 3x$.



(a) Find the value of x.

$x =$ ..... **[2]**

(b) Event A is that the spinner lands on an even number.
Find

(i) $P(A)$,

$P(A) =$ ..... **[2]**

(ii) $P(A')$.

$P(A') =$ ..... **[1]**

(c) The spinner is spun 200 times.
Calculate an estimate for the number of times the spinner lands on a prime number.

..... **[2]**

6 A biased spinner can land on 1, 2, 3 or 4.
The table shows the probabilities of the spinner landing on each number.

Number	1	2	3	4
Probability	$\frac{1}{5}$	0.35	x	$2x$

(a) Work out the value of x.

$x =$ ..... **[2]**

(b) Find the probability that the spinner does not land on 4.

..... **[2]**

7 A biased four-sided die is rolled.
The possible scores are 1, 2, 3 and 4.
The table shows the probabilities of rolling each score.

Score	1	2	3	4
Probability	p^2	p	0.15	0.1

(a) Show that $4p^2 + 4p - 3 = 0$.

[2]

(b) Find the value of p.

$p =$ ..... **[2]**

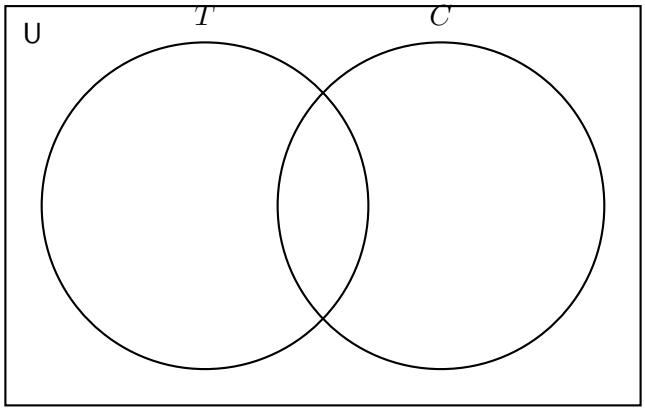
(c) Let E be the event of rolling an even number.
Work out $P(E')$.

$P(E') =$ ..... **[2]**

8 In a group of 60 people, some like tea (T) and some like coffee (C).

- The total number of people who like tea is 38.
- The total number of people who like coffee is 24.
- The number of people who like neither is x.
- The number of people who like both is $2x$.

(a) Use this information to complete the Venn diagram in terms of x.



[3]

(b) Find the value of x.

$x =$ ..... **[2]**

(c) A person is chosen at random from the group.
Find $P(T \cap C')$.

$P(T \cap C') =$ ..... **[2]**

9 A bag contains only red counters, blue counters, and green counters.
The number of red counters is x^2.
The number of blue counters is $3x$.
The number of green counters is 2.

(a) Write down an expression, in its simplest form, for the total number of counters in the bag.

..... **[1]**

(b) A counter is chosen at random.
The probability that it is red is $\frac{8}{15}$.
Show that $7x^2 - 24x - 16 = 0$.

[2]

(c) Solve the equation $7x^2 - 24x - 16 = 0$ to find the value of x.

$x =$ ..... **[3]**

(d) Find the probability that a randomly chosen counter is not blue.
Give your answer as a fraction in its simplest form.

..... **[2]**

10 A box contains x red counters, $(2x-1)$ blue counters, and (x^2-4) green counters, where $x > 2$.
A counter is taken at random from the box.
Event R is taking a red counter, and event G is taking a green counter.
It is given that $P(G) = \frac{3}{5}$.

(a) Show that $2x^2 - 9x - 5 = 0$.

[3]

(b) Solve the equation $2x^2 - 9x - 5 = 0$ to find the number of red counters in the box.

$x =$ ..... **[2]**

(c) Find, as a fraction in its simplest form,

(i) $P(R)$,

$P(R) =$ ..... **[2]**

(ii) $P(G')$.

$P(G') =$ ..... **[2]**

(d) A further k red counters are added to the box.
When a counter is now taken at random from the box, $P(R') = \frac{3}{4}$.
Find the value of k.

$k =$ ..... **[3]**

11 A bag contains only red marbles, blue marbles, and green marbles.
The probability of choosing a red marble at random is $\frac{2}{5}$.
The probability of choosing a blue marble at random is $\frac{1}{3}$.
There are 12 green marbles in the bag.

(a) Calculate the probability of choosing a green marble at random.

..... **[2]**

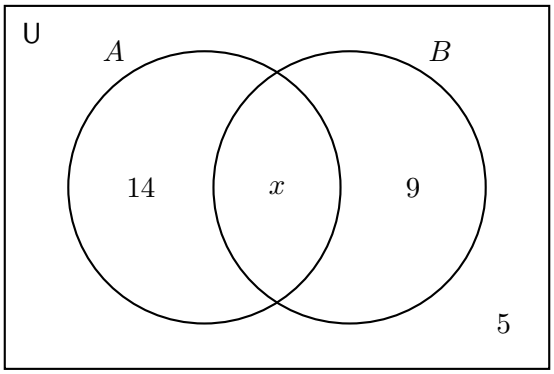
(b) Calculate the total number of marbles in the bag.

..... **[3]**

(c) Write down the probability that a marble chosen at random is not blue.

..... **[1]**

12 The Venn diagram shows the number of elements in the universal set U and sets A and B.
There are 36 elements in total in the universal set U.



(a) Find the value of x.

$x =$ ..... **[2]**

(b) An element is chosen at random from the universal set.
Find $P(A)$.

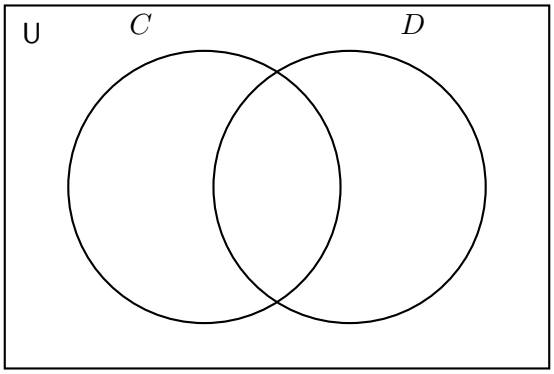
$P(A) =$ ..... **[2]**

(c) Find $P(B')$.

$P(B') =$ ..... **[2]**

13 A group of 120 people were asked about the pets they own.
$C = \{\text{people who own a cat}\}$
$D = \{\text{people who own a dog}\}$
$n(\mathsf{U}) = 120$
$n(C) = 55$
$n(D) = 68$
18 people own neither a cat nor a dog.

(a) Complete the Venn diagram to show the number of people in each region.



[3]

(b) A person is chosen at random.
Find $P(C \cap D)$.

..... **[1]**

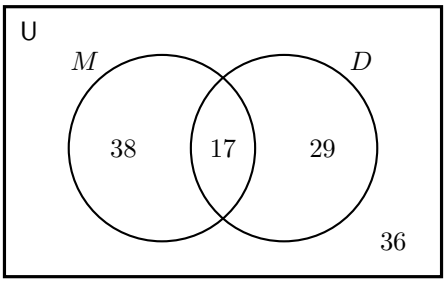
(c) Find $P(D')$.

..... **[2]**

(d) Find the probability that a person chosen at random owns a cat or a dog, but not both.

..... **[2]**

14 In a survey of 120 students, M is the set of students who study Music and D is the set of students who study Drama.
The Venn diagram shows the number of students in each subset, where U is the set of all 120 students.



(a) Write down $n((M \cup D)')$.

..... **[1]**

(b) A student is chosen at random from the 120 students.
Find, as a fraction in its simplest form,

(i) $P(M)$,

$P(M) =$ ..... **[2]**

(ii) $P(M' \cap D)$,

$P(M' \cap D) =$ ..... **[2]**

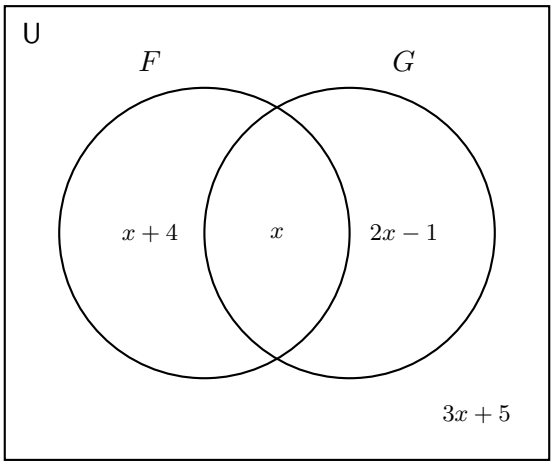
(iii) $P((M \cap D)')$.

$P((M \cap D)') =$ ..... **[2]**

(c) One of the students who studies Music is chosen at random.
Write down the probability that this student also studies Drama.

..... **[2]**

15 The Venn diagram shows information about the number of students in a class of 50 who study French (F) and German (G).



(a) Work out the value of x.

$x =$ ..... **[2]**

(b) A student is chosen at random.
Find $P(F')$.

..... **[2]**

(c) A student is chosen at random.
Find the probability that this student studies exactly one of these languages.

..... **[2]**

1 **(a)** 0.17 **[B1]**

(b) The train does not arrive on time (or equivalent) **[B1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** $1 - \left(\frac{2}{7} + \frac{1}{3}\right)$ or equivalent **[M1]**

$\frac{8}{21}$ **[A1]**

(b) $\frac{5}{7}$ **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** $0.15 + 2x + 0.31 + x + 0.24 = 1$ or $3x + 0.7 = 1$ **[M1]**

0.1 **[A1]**

(b) $P(2) = 2(0.1) = 0.2$ **[M1]**

0.8 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Sum of given probabilities: $0.15 + 0.3 + 0.45 = 0.9$ **[M1]**

0.1 **[A1]**

(b) 0.45 **[B1]**

(c) Uses $1 - P(G)$ formula: $1 - 0.45$ **[M1]**

0.55 **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $0.15 + 2x + 0.35 + 3x = 1$ **[M1]**

$5x = 0.50$, so $x = 0.1$ (or 0.10) **[A1]**

(b) **(i)** $P(A) = P(2) + P(4) = 0.20 + 0.30$ **[M1]**

0.5 (or 0.50 or $\frac{1}{2}$) **[A1]**

(ii) $1 - 0.5 = 0.5$ (or 0.50 or $\frac{1}{2}$) **[B1]**

(c) $P(2) + P(3) = 0.20 + 0.35 = 0.55$ **[M1]**

$200 \times 0.55 = 110$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $\frac{1}{5} + 0.35 + x + 2x = 1$ or $0.55 + 3x = 1$ **[M1]**

0.15 **[A1]**

(b) $1 - 2(0.15)$ or $\frac{1}{5} + 0.35 + 0.15$ **[M1]**

0.7 or $\frac{7}{10}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

7 **(a)** $p^2 + p + 0.15 + 0.1 = 1$ **[M1]**

$p^2 + p - 0.75 = 0$ leading to $4p^2 + 4p - 3 = 0$ with no errors seen **[A1]**

(b) $(2p - 1)(2p + 3) = 0$ or correct substitution into the quadratic formula **[M1]**

0.5 (or $\frac{1}{2}$) **[A1]**

(c) $P(E) = 0.5 + 0.1 = 0.6$ or $P(E') = P(1) + P(3)$ **[M1]**

0.4 (or $\frac{2}{5}$) **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** Places $2x$ in the intersection **[B1]**

Places $38 - 2x$ in T only region and $24 - 2x$ in C only region **[B1]**

Places x in the region outside both circles **[B1]**

(b) Forms correct equation: $(38 - 2x) + 2x + (24 - 2x) + x = 60$ or $62 - x = 60$ **[M1]**

2 **[A1]**

(c) Finds the number of people in $T \cap C'$: $38 - 2(2) = 34$ **[M1]**

$\frac{34}{60}$ or $\frac{17}{30}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** $x^2 + 3x + 2$ **[B1]**

(b) $\frac{x^2}{x^2 + 3x + 2} = \frac{8}{15}$ **[M1]**

$15x^2 = 8(x^2 + 3x + 2) \implies 15x^2 = 8x^2 + 24x + 16$ **[M1]**

(c) $(7x + 4)(x - 4) = 0$ or correct substitution into the quadratic formula [M2]
(Award M1 for $(7x + a)(x + b) = 0$ where $ab = -16$ or $7b + a = -24$)

4 **[A1]**

(d) Total counters $= 4^2 + 3(4) + 2 = 30$ and number of blue counters $= 3(4) = 12$ **[M1]**

$\frac{18}{30} = \frac{3}{5}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

10 **(a)** Total number of counters $= x + (2x - 1) + (x^2 - 4) = x^2 + 3x - 5$ **[B1]**

$\dfrac{x^2 - 4}{x^2 + 3x - 5} = \dfrac{3}{5}$ giving $5(x^2 - 4) = 3(x^2 + 3x - 5)$ **[M1]**

$5x^2 - 20 = 3x^2 + 9x - 15$ simplifying to $2x^2 - 9x - 5 = 0$ **[A1]**

(b) $(2x + 1)(x - 5) = 0$ **[M1]**

5 (rejecting $x = -\dfrac{1}{2}$ since $x > 2$) **[A1]**

(c) **(i)** Total counters $= 5 + 9 + 21 = 35$, giving $P(R) = \dfrac{5}{35}$ **[M1]**

$\dfrac{1}{7}$ **[A1]**

(ii) $P(G') = 1 - P(G) = 1 - \dfrac{3}{5}$ **[M1]**

$\dfrac{2}{5}$ **[A1]**

(d) Number of counters that are not red $= 9 + 21 = 30$, giving $\dfrac{30}{35 + k} = \dfrac{3}{4}$ **[M1]**

$3(35 + k) = 120$ giving $35 + k = 40$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 12]*

11 **(a)** $1 - \left(\dfrac{2}{5} + \dfrac{1}{3}\right)$ or $1 - \dfrac{11}{15}$ **[M1]**

$\dfrac{4}{15}$ **[A1]**

(b) Equates their fraction to the number of green marbles: $\dfrac{4}{15}$ of total $= 12$ **[M1]**

Total $= 12 \times \dfrac{15}{4}$ **[M1]**

45 **[A1]**

(c) $\dfrac{2}{3}$ **[B1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** $14 + x + 9 + 5 = 36$ or equivalent **[M1]**

8 **[A1]**

(b) $\dfrac{14 + 8}{36}$ or $\dfrac{22}{36}$ **[M1]**

$\dfrac{11}{18}$ **[A1]**

(c) $1 - P(B)$ or $\dfrac{14 + 5}{36}$ **[M1]**

$\dfrac{19}{36}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** 21 in the intersection $C \cap D$ **[B1]**

34 in C only and 47 in D only **[B1]**

18 outside both circles **[B1]**

(b) $\frac{21}{120}$ or $\frac{7}{40}$ **[B1]**

(c) $1 - \frac{68}{120}$ or $\frac{34+18}{120}$ **[M1]**

$\frac{52}{120}$ or $\frac{13}{30}$ **[A1]**

(d) $\frac{34+47}{120}$ or $\frac{120-21-18}{120}$ **[M1]**

$\frac{81}{120}$ or $\frac{27}{40}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

14 **(a)** 36 **[B1]**

(b) **(i)** $n(M) = 38 + 17 = 55$, giving $\frac{55}{120}$ **[M1]**

$\frac{11}{24}$ **[A1]**

(ii) $n(M' \cap D) = 29$ **[B1]**

$\frac{29}{120}$ **[B1]**

(iii) $1 - \frac{17}{120}$ (or $\frac{38+29+36}{120}$) **[M1]**

$\frac{103}{120}$ **[A1]**

(c) Total studying Music is 55, and 17 also study Drama **[M1]**

$\frac{17}{55}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

15 **(a)** $(x+4) + x + (2x-1) + (3x+5) = 50$ or $7x + 8 = 50$ **[M1]**

6 **[A1]**

(b) $(2(6)-1) + (3(6)+5)$ or $50 - (6+4+6)$ or $\frac{11+23}{50}$ **[M1]**

$\frac{34}{50}$ or $\frac{17}{25}$ or 0.68 **[A1]**

(c) $(6+4) + (2(6)-1)$ or $10 + 11$ or $\frac{21}{50}$ **[M1]**

$\frac{21}{50}$ or 0.42 **[A1]**

[Back to Question] *[Total Marks: 6]*